

Python Assignment02

Level: Intermediate | Total Questions: 20

Instructions: Write a Python program for each question below. Test your code with at least two different inputs before submitting.

Section 1: List

1. Create a list of 5 numbers and print the sum of all the numbers in the list.
Hint: Use a loop with a variable like total = 0, and add each number to it.
2. Create a list of 5 fruit names and print only the first and last fruit from the list.
Hint: Remember list index starts from 0, and the last item can be accessed with index -1.
3. Write a program that takes a list of numbers and counts how many of them are greater than 10.
Hint: Use a loop, an if condition, and a counter variable starting at 0.
4. Write a program to add a new item to a list, then remove one item from it, and print the list after each step.
Hint: Use list.append() to add an item and list.remove() to delete an item.

Section 2: String

5. Take a string from the user and print its total number of characters (length).
Hint: Use the len() function on the string.
6. Take a string from the user and print it in all uppercase and all lowercase.
Hint: Use the .upper() and .lower() string methods.
7. Write a program that checks whether a given word is present inside a sentence (string).
Hint: Use the 'in' keyword, like: if word in sentence:

Section 3: Loop

8. Write a program to print numbers from 1 to 20 using a loop.
Hint: Use range(1, 21) with a for loop.
9. Write a program to print only the even numbers from 1 to 30 using a loop.
Hint: Inside the loop, use an if condition to check number % 2 == 0.
10. Write a program to find the sum of numbers from 1 to 50 using a loop.
Hint: Start with total = 0 before the loop, then add each number to total inside the loop.

Section 4: Function

11. Write a function named `greet(name)` that takes a name and prints 'Hello, <name>!'. Call the function with two different names.
Hint: Define the function with `def greet(name):`, then call it like `greet('Rahim')`.
12. Write a function named `add_numbers(a, b)` that takes two numbers and returns their sum. Print the result by calling the function.
Hint: Use the 'return' keyword to send the result back, then print the function call: `print(add_numbers(3, 5))`.
13. Write a function named `square(number)` that returns the square of a given number. Call it with three different numbers.
Hint: Square of a number means `number * number`.

Section 5: Nested if

14. Write a program that takes a number and checks whether it is positive or negative. If it is positive, check whether it is even or odd using a nested if.
Hint: First write the outer if-else for positive/negative, then put another if-else inside the 'positive' block to check even/odd.
15. Write a program that takes a student's marks and first checks if the marks are above or below 40 (pass/fail). If passed, use a nested if to decide the grade (A, B, or C).
Hint: Outer if checks marks ≥ 40 for pass/fail. Inside the pass block, use nested if-elif-else for grade ranges.
16. Write a program that takes a person's age. First check if the age is above 18 or not. If above 18, use a nested if to check if the age is above 60 as well (senior citizen).
Hint: Outer if checks age > 18 . Inside that block, add another if to check age > 60 .

Section 6: Loop in Function

17. Write a function named `print_numbers(n)` that uses a loop inside the function to print numbers from 1 to n.
Hint: Put a for loop with `range(1, n+1)` inside the function body.
18. Write a function named `count_down(n)` that uses a loop inside the function to print numbers from n down to 1.
Hint: Use `range(n, 0, -1)` to count downward.
19. Write a function named `sum_of_list(numbers)` that takes a list and uses a loop inside the function to calculate and return the total sum of the list.
Hint: Initialize `total = 0` inside the function, loop through the list adding each item, then return total.
20. Write a function named `print_table(number)` that uses a loop inside the function to print the multiplication table of that number from 1 to 10.
Hint: Loop from 1 to 10, and inside the loop print `number * i` for each i.